

13.4 Distance Metrics



Core skills you will master in this section:

- ▶ Mahalanobis distance: measures the distance from a sample point to the mean, taking into account the variance and covariance of the features.
- ▶ In the plane, the Mahalanobis distance's contour lines are rotated ellipses.
- ▶ The unit of Mahalanobis distance is the standard deviation.
- ▶ The geometric transformation behind Mahalanobis distance: translation → rotation → scaling.
- ▶ Standardized Euclidean distance: eliminates the effects of variable scale and units, but does not account for correlation.
- ▶ In the plane, the standardized Euclidean distance's contour lines are axis-aligned ellipses.
- ▶ In the plane, the Euclidean distance's contour lines are circles.

This section examines several common distance metrics from the perspective of quadratic forms; pay particular attention to the connections and differences between Mahalanobis distance and the other distances. You have frequently seen data ellipses earlier in this book — the Mahalanobis distance in this section will explain the mathematical principle behind these rotated ellipses.

Mahalanobis Distance

As shown in Figure 1, A, B, C, D all lie on the same circle centered at the data's centroid. Clearly, A, B, C, D are all the same straight-line distance (Euclidean distance) from the data centroid μ .

But looking at how tightly the "data cloud" is distributed, C and D are "closer" to the data centroid μ , while A and B are "farther" from μ .

Mahalanobis distance (Mahalanobis distance, Mahal distance) is exactly the distance metric that accounts for this kind of distribution. Mahalanobis distance is defined as

$$d_{\text{Mahal}}(\mathbf{x}, \mu) = \sqrt{(\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu)}$$

where $\mathbf{x}$ represents the sample column vector, μ is the data centroid vector, and Σ is the covariance matrix.

 Have you noticed that the expression above also appears in the probability density function of the multivariate Gaussian distribution?

 Note that the covariance matrix Σ must be invertible.

Mahalanobis distance measures how far a sample point $\mathbf{x}$ deviates from the overall mean μ , taking into account both the dispersion of each variable and the linear correlations among the variables.

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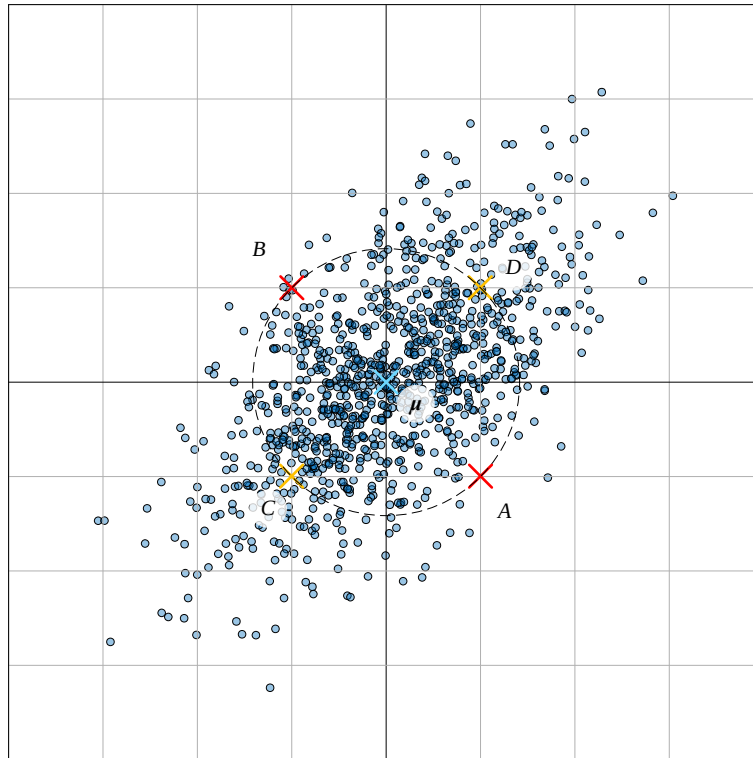


Figure 1. Near and Far, from the Perspective of Mahalanobis Distance

The square of the Mahalanobis distance is

$$d_{\text{Mahal}}^2(\mathbf{x}, \boldsymbol{\mu}) = (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})$$

The equation above is essentially a quadratic form.

Let's look at an example.

Given the following covariance matrix $\boldsymbol{\Sigma}$

$$\boldsymbol{\Sigma} = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1 \end{bmatrix}$$

The inverse of this covariance matrix is then

$$\boldsymbol{\Sigma}^{-1} = \begin{bmatrix} 4/3 & -2/3 \\ -2/3 & 4/3 \end{bmatrix}$$

Suppose the data's centroid $\boldsymbol{\mu}$ is

$$\boldsymbol{\mu} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Figure 1 — point A's corresponding column vector is

$$\mathbf{x}_A = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Compute the Mahalanobis distance between point A and the centroid μ

$$(x - \mu)^T \Sigma^{-1} (x - \mu) = \left(\begin{bmatrix} 1 \\ -1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right)^T @ \begin{bmatrix} 4/3 & -2/3 \\ -2/3 & 4/3 \end{bmatrix} @ \left(\begin{bmatrix} 1 \\ -1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right) = 4$$

Taking the square root gives a Mahalanobis distance of 2 between point A and the centroid μ .



Please compute the Mahalanobis distance from point B, $[-1; 1]$, to μ .

Figure 1 — point C's corresponding column vector is

$$x_C = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

Compute the Mahalanobis distance between point C and the centroid μ

$$(x - \mu)^T \Sigma^{-1} (x - \mu) = \left(\begin{bmatrix} -1 \\ -1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right)^T @ \begin{bmatrix} 4/3 & -2/3 \\ -2/3 & 4/3 \end{bmatrix} @ \left(\begin{bmatrix} -1 \\ -1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right) = \frac{4}{3}$$

Taking the square root gives a Mahalanobis distance of about 1.155 between point C and the centroid μ .



Please compute the Mahalanobis distance from point D, $[1; 1]$, to μ .

Mahalanobis Distance Contours: Rotated Ellipses

In the equation above, the covariance matrix Σ 's corresponding Mahalanobis distance contours are shown in Figure 2. What we see is a set of concentric rotated ellipses — this ellipse is exactly the source of the rotated ellipses you've seen earlier in this book. In the figure, points A and B lie on the contour where the Mahalanobis distance equals 2; points C and D have a Mahalanobis distance slightly greater than 1.

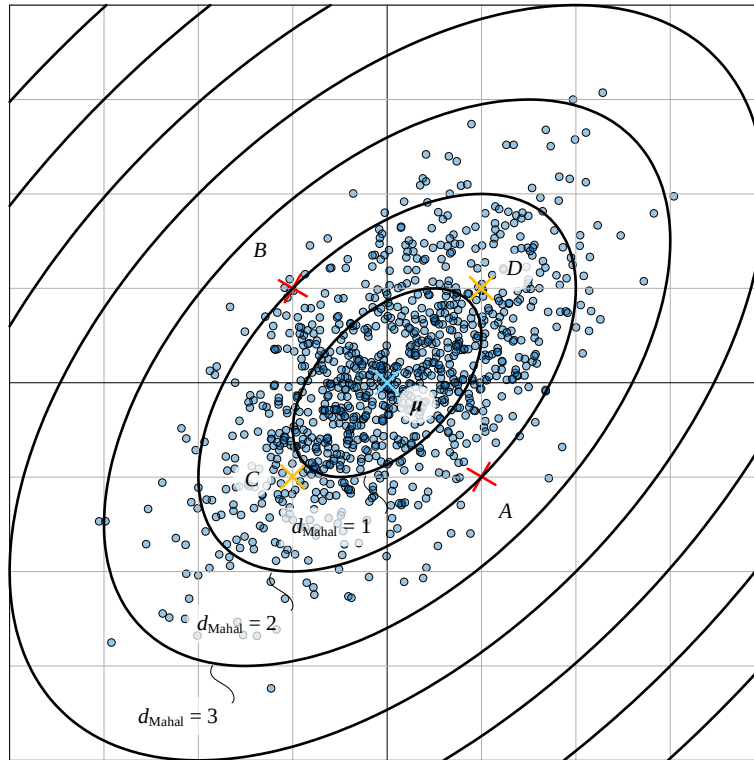


Figure 2. Mahalanobis Distance Contours

⚠ Note that the unit of Mahalanobis distance is the "standard deviation". For example, if the computed Mahalanobis distance is 2, it should be referred to as 2 standard deviations.

Eigenvalue Decomposition: Scaling → Rotation → Translation

Σ spectrally decomposed gives:

$$\Sigma = V\Lambda V^T$$

where V is an orthogonal matrix.

Σ^{-1} 's eigenvalue decomposition can be written as:

$$\Sigma^{-1} = V\Lambda^{-1}V^T$$

Substituting the equation above into the previous equation *gives*:

$$\begin{aligned} d_{\text{Mahal}}^2(\mathbf{x}, \boldsymbol{\mu}) &= (\mathbf{x} - \boldsymbol{\mu})^T V\Lambda^{-1}V^T (\mathbf{x} - \boldsymbol{\mu}) \\ &= (\mathbf{x} - \boldsymbol{\mu})^T V\Lambda^{-\frac{1}{2}}\Lambda^{-\frac{1}{2}}V^T (\mathbf{x} - \boldsymbol{\mu}) \\ &= \left(\Lambda^{-\frac{1}{2}}V^T (\mathbf{x} - \boldsymbol{\mu}) \right)^T \left(\Lambda^{-\frac{1}{2}}V^T (\mathbf{x} - \boldsymbol{\mu}) \right) \\ &= \left\| \Lambda^{-\frac{1}{2}}V^T (\mathbf{x} - \boldsymbol{\mu}) \right\|^2 \end{aligned}$$

Taking the square root gives

$$d_{\text{Mahal}}(\mathbf{x}, \boldsymbol{\mu}) = \left\| \underbrace{\mathbf{A}^{-\frac{1}{2}} \mathbf{V}^T}_{\substack{\text{Scale} \quad \text{Rotate}}} (\mathbf{x} - \boldsymbol{\mu}) \right\|$$

Let

$$\mathbf{z} = \mathbf{A}^{-\frac{1}{2}} \mathbf{V}^T (\mathbf{x} - \boldsymbol{\mu})$$

The equation above means that the column vector $\mathbf{x}$ is first translated using $-\boldsymbol{\mu}$, and then rotated (about the origin) using the $\mathbf{V}$ matrix; finally, $\mathbf{A}^{-\frac{1}{2}}$ completes the scaling. The full geometric transformation process is shown in Figure 3.

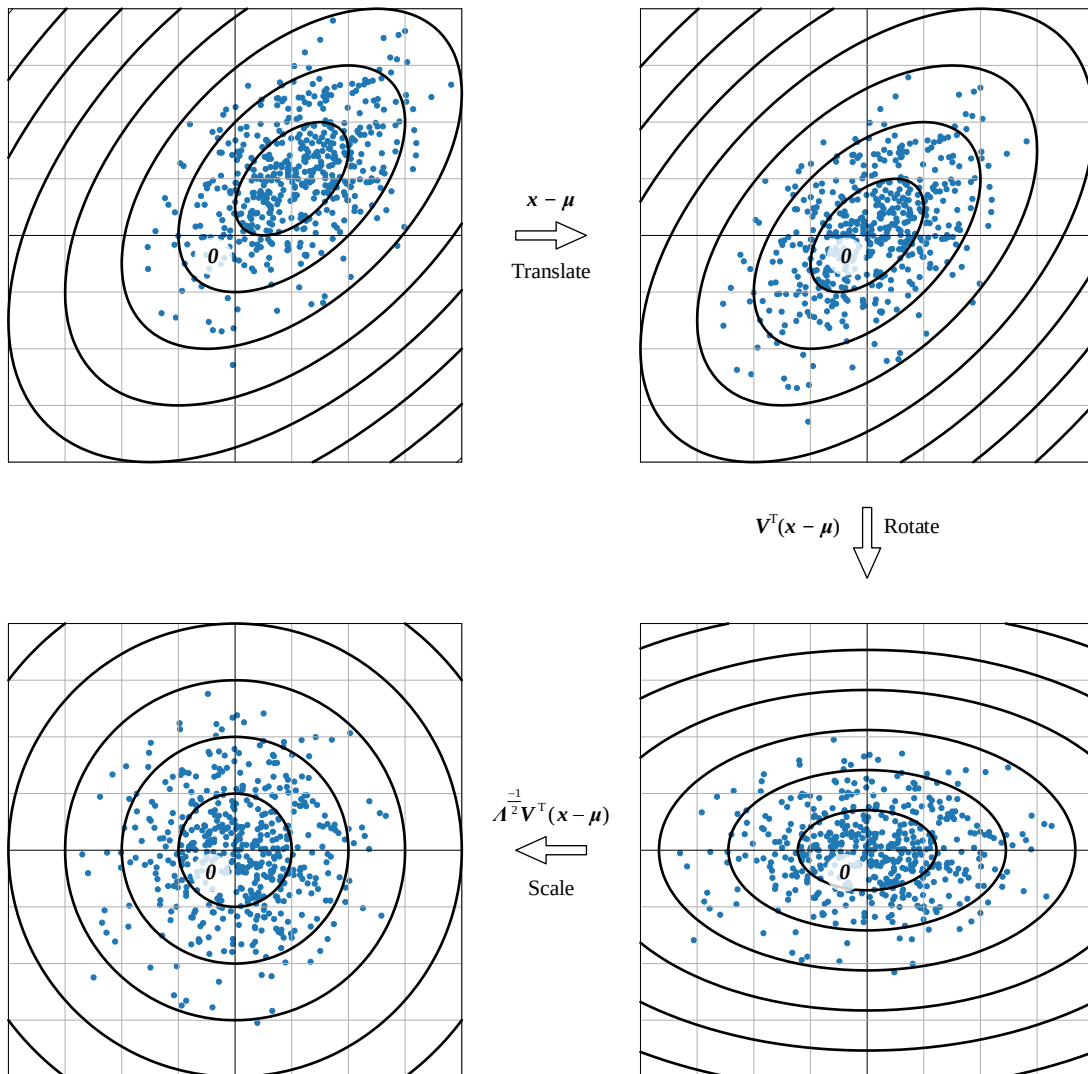


Figure 3. Mahalanobis Distance Contours: "Translation → Rotation → Scaling"

Figure 3 — looking at this in reverse, the grid for Euclidean distance is a unit square, and after $\mathbf{A}^{\frac{1}{2}} \mathbf{V}$ the "inverse" linear transformation, the unit-square grid becomes a rotated rectangle, as shown specifically in Figure 4.

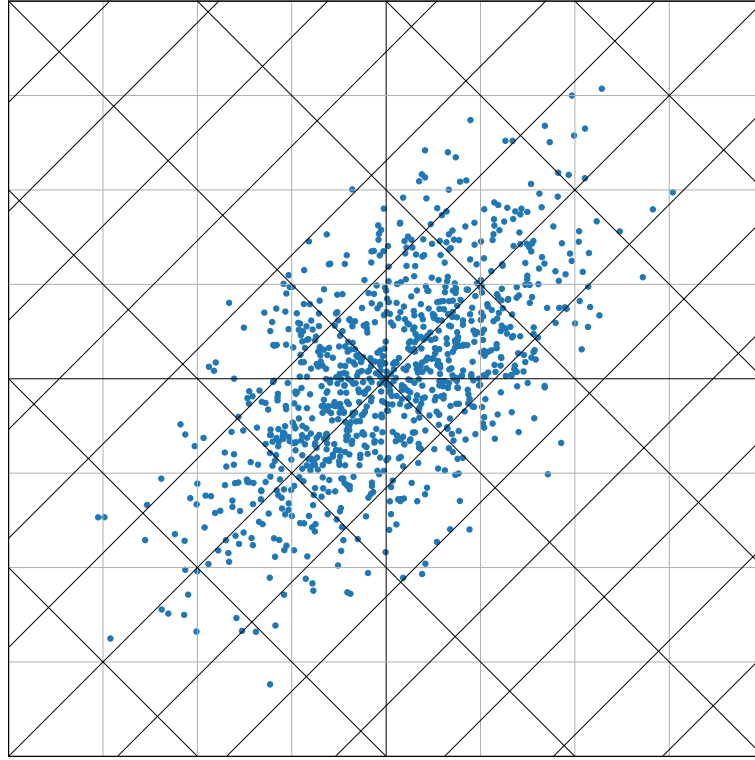


Figure 4. Mahalanobis Distance Grid

Standardized Euclidean Distance

Replacing the covariance matrix Σ with the following diagonal matrix gives

$$\mathbf{D}^2 = \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_D^2 \end{bmatrix}$$

This gives us another distance metric – standardized Euclidean distance, namely

$$d(\mathbf{x}, \boldsymbol{\mu}) = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^T \mathbf{D}^{-2} (\mathbf{x} - \boldsymbol{\mu})}$$

Standardized Euclidean distance eliminates the differences in measurement units and scale among different features, thereby reducing bias in the distance computation.

Unlike Euclidean distance, standardized Euclidean distance does not account for the correlations among the data's different features. This is also reflected in the equation above – this matrix is a diagonal matrix, whose main-diagonal elements are the variances of each feature, and whose off-diagonal elements are 0.

For $D = 2$, the two-feature case, the standardized Euclidean distance between a sample point $\mathbf{x}$ and the centroid $\boldsymbol{\mu}$ can be written as:

$$d = \sqrt{\left(\frac{x_1 - \mu_1}{\sigma_1}\right)^2 + \left(\frac{x_2 - \mu_2}{\sigma_2}\right)^2}$$

As shown in Figure 5, the standardized Euclidean distance contours are axis-aligned ellipses.

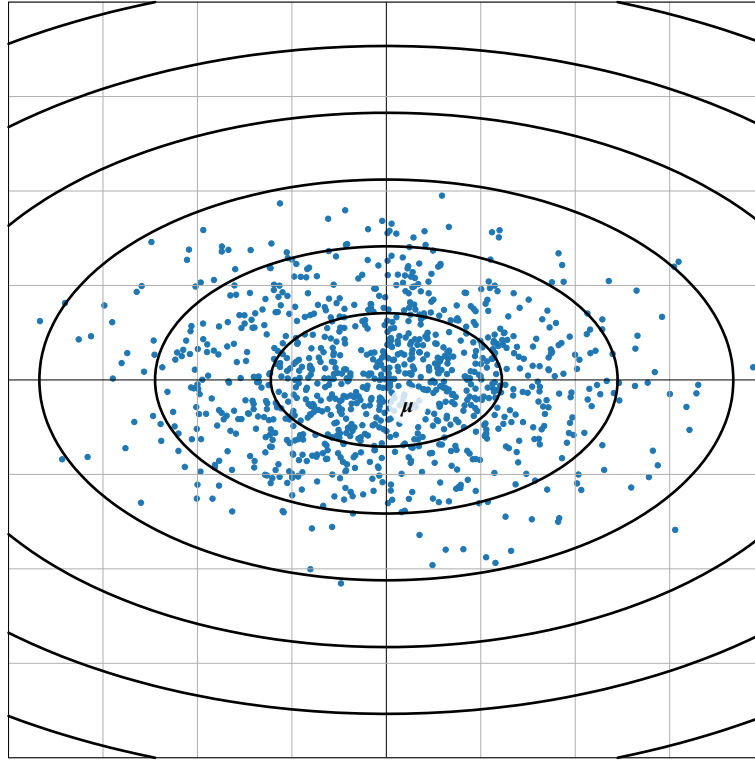


Figure 5. Standardized Euclidean Distance Contours

Euclidean Distance

Replacing the covariance matrix in the equation above with the identity matrix gives us the Euclidean distance

$$d(\mathbf{x}, \boldsymbol{\mu}) = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^T \mathbf{I}^{-1} (\mathbf{x} - \boldsymbol{\mu})} = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^T (\mathbf{x} - \boldsymbol{\mu})}$$

For $D = 2$, the two-feature case, the Euclidean distance between a sample point $\mathbf{x}$ and the centroid $\boldsymbol{\mu}$ can be written as:

$$d = \sqrt{(x_1 - \mu_1)^2 + (x_2 - \mu_2)^2}$$

As shown in Figure 6, the Euclidean distance contours are circles.

Euclidean distance is an intuitive geometric distance; it has units, depends directly on the measurement units of the raw data, treats differences between variables equally, and does not account for correlations among variables. As a result, Euclidean distance can become distorted when different variables have very different units or variances.

In contrast, Mahalanobis distance is a dimensionless distance metric: by introducing the covariance matrix, it comprehensively accounts for the linear correlations among variables, thereby eliminating redundant information and the interference of inter-variable correlation, making the resulting distance more statistically meaningful.

Mahalanobis distance can also effectively identify outliers — the larger its value, the more a sample deviates from the overall distribution — but its use assumes the variables approximately follow a multivariate normal distribution.



We mentioned earlier in this book that Euclidean distance is a special case of the L^p norm. Please review the concept of the L^p norm, especially the cases $p = 1, 2, \infty$.

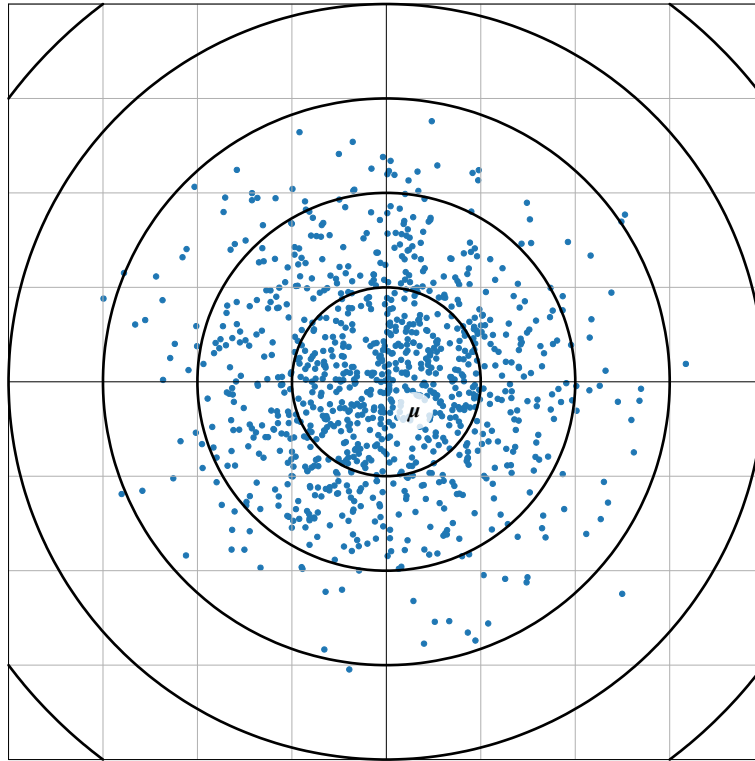


Figure 6. Euclidean Distance Contours



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Please learn to use the following function to compute the Mahalanobis distance.

<https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.distance.mahalanobis.html>

Q2. Please learn to use the following function to compute the standardized Euclidean distance.

<https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.distance.seuclidean.html>

Q3. Please learn about the Manhattan distance (city block distance), and learn to use the following function

<https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.distance.cityblock.html>

Q4. Please learn about the Chebyshev distance, and learn to use the following function

<https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.distance.chebyshev.html>

Q5. Please learn about the Minkowski distance, and learn to use the following function

<https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.distance.minkowski.html>

Q6. Please learn about the cosine distance, and learn to use the following function

<https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.distance.cosine.html>